

Emergent Geometry and the Structure of Matter and Time

Chase Deggs*

Abstract

We develop a geometric framework in which the structure of matter and the orientation of time arise from a single underlying manifold construction. The model begins with a substrate field whose density seeds proto-geometric structure and generates an intermediate dark-sector metric, from which the observable spacetime geometry emerges through an excitation boundary operator acting on matter fields. A lemniscate foliation of the manifold, defined by a scalar operator $\mathcal{F}$, introduces a global temporal orientation, a self-intersection locus, and a natural partitioning of spacetime into causally distinct sectors. Excitations satisfying closed boundary conditions correspond to baryonic matter, while open boundaries propagate as non-baryonic modes whose internal evolution is governed by a classical $SU(3)$ transport geometry. Curvature, torsion, and twist on this internal manifold contribute directly to flavor dynamics, producing geometric analogs of the atmospheric, solar, and CP-phase sectors. In this picture, matter identity, flavor evolution, and temporal directionality are not independent inputs but emergent consequences of the underlying geometry.

The resulting framework links metric hierarchy, foliation structure, and internal flavor transport into a unified geometric description that suggests phenomena typically treated as separate—curvature, causal structure, and flavor mixing—may share a common geometric origin. By incorporating matter identity and temporal orientation into the same geometric hierarchy that produces curvature, the model extends emergent-geometry programs and provides a route to observational consequences, including geometric corrections to neutrino oscillations, dark-sector-induced anisotropies, and large-scale structure signatures. These features position the framework as a geometric unification of matter identity, temporal directionality, and flavor transport.

*Independent Researcher

1 Introduction

The search for a unified description of matter, geometry, and temporal structure remains a central goal of theoretical physics. Conventional approaches treat spacetime geometry as fundamental, with matter fields propagating on a pre-existing metric background. Emergent-gravity programs invert this hierarchy, proposing that curvature, torsion, and metric structure arise from deeper degrees of freedom [1, 2]. These approaches offer conceptual advantages, yet they typically address only isolated aspects of the problem: geometry without matter identity, matter without geometric origin, or dark-sector behavior without connection to baryonic structure.

This work develops a geometric framework in which matter identity, flavor evolution, and temporal orientation arise from a single underlying manifold construction. Rather than competing with existing emergent-geometry programs, the present framework extends them by incorporating matter identity and temporal orientation into the same geometric hierarchy that produces curvature. The model is built on three interacting fields:

1. A substrate density field Φ_{sub} whose gradient flow seeds proto-geometric structure. Its integral curves form filamentary networks reminiscent of cosmic-web formation [3], providing the scaffolding from which curvature emerges.
2. A dark-sector phase field Φ_{dark} that responds to substrate density through conformal–disformal coupling [4, 5]. This field generates an intermediate metric $g_{\mu\nu}^{\text{dark}}$ that biases the emergent geometry and introduces anisotropic responses relevant to cosmology.
3. A universal excitation field Ψ whose geometric boundary operator $L[\Psi]$ determines matter identity. Excitations satisfying $L[\Psi] = 0$ form closed geometric boundaries analogous to topological solitons [6], corresponding to baryonic matter. Excitations lacking closed boundaries propagate through an internal $\text{SU}(3)$ manifold, treated here as a classical geometric space whose curvature, torsion, and twist give rise to effective flavor evolution [7, 8].

A key addition of the present framework is a lemniscate foliation of the manifold, defined by a scalar operator $\mathcal{F}$ whose level sets form one-dimensional leaves. The foliation introduces a global temporal orientation, a self-intersection locus Σ , and a natural partitioning of spacetime into causally distinct sectors. Temporal directionality arises from the requirement that the orientation field τ satisfies $D\mathcal{F}(\tau) > 0$, ensuring monotonic flow across the foliation and enforcing causal consistency.

These elements combine to produce a fully emergent spacetime metric $g_{\mu\nu}$. The substrate field establishes geometric scaffolding, the dark-sector field modifies metric response, the foliation imposes temporal orientation, and the excitation field determines matter identity and flavor transport. In this picture, curvature, torsion, twist, matter identity, and temporal directionality are not independent inputs but emergent consequences of the underlying geometry.

The internal $SU(3)$ manifold governing non-baryonic excitations defines a geometric Hamiltonian H_{geom} whose curvature, torsion, and twist map onto the atmospheric, solar, and CP-phase sectors of the PMNS matrix. This mapping provides a direct route to experimental tests, including potential deviations from the MSW effect [9, 10] in long-baseline experiments such as DUNE.

The model also predicts dark-sector-induced anisotropies in the cosmic microwave background, filament alignment scales determined by substrate gradients, and structure-formation deviations arising from disformal coupling. These predictions offer observational pathways for distinguishing emergent geometry from modified-gravity or dark-energy models.

To support readers in navigating the geometric and physical structure of the model, we include a comprehensive set of reference charts summarizing metric hierarchy, geometric operators, $SU(3)$ generator roles, field interactions, and experimental signatures. These charts appear before the bibliography and serve as a conceptual guide to the framework.

The remainder of the paper is organized as follows. Section 2 introduces foundational operators and terminology. Sections 3–5 develop the substrate, dark-sector, and excitation fields. Section 6 presents the lemniscate foliation and temporal orientation. Section 7 develops the $SU(3)$ geometric manifold and neutrino transport. Section 8 maps geometric quantities to PMNS parameters. Section 9 compares the model to existing $SU(3)$ compactification and geometric-phase frameworks. Section 10 presents the unified action and field equations. Section 11 details observational signatures. Section 12 provides reference charts summarizing key relationships. Section 13 contains the bibliography.

2 Terminology and Foundational Operators

This section establishes the terminology, geometric structures, and foundational operators used throughout the emergent-geometry framework. Because the model integrates substrate dynamics, dark-sector phase response, geometric boundary conditions, and internal $SU(3)$ transport, it is essential to define the mathematical objects that appear repeatedly in later sections. Reference Charts at the end of the paper summarize these operators and their physical interpretations.

2.1 Emergent Metric Hierarchy

A central feature of the framework is the existence of three distinct metrics, each arising from different levels of field interaction. These metrics form a hierarchy that reflects the emergent nature of spacetime geometry in the model.

- **Reference Metric g_0 :** A fixed background metric used solely for variational formulation. It carries no physical dynamics and serves as the baseline from which emergent geometry is constructed.
- **Dark-Sector Metric $g_{\mu\nu}^{\text{dark}}$:** Generated by the conformal–disformal response of the dark-sector phase field Φ_{dark} . This metric introduces anisotropic and density-dependent modifications to geometric structure. Its definition follows the general disformal transformation introduced by Bekenstein [4]:

$$g_{\mu\nu}^{\text{dark}} = k(\Phi_{\text{dark}}) g_{0\mu\nu} + h(\Phi_{\text{dark}}) \partial_\mu \Phi_{\text{dark}} \partial_\nu \Phi_{\text{dark}}. \quad (1)$$

- **Emergent Metric $g_{\mu\nu}$:** The fully dynamical spacetime metric arising from the combined influence of Φ_{sub} , Φ_{dark} , and Ψ . This metric governs all physical propagation and curvature in the model.

The relationships among these metrics are summarized in the Metric Hierarchy Chart in the Reference Charts section.

2.2 Lemniscate Boundary Operator

Matter identity in the framework is determined by the geometric boundary operator $L[\Psi]$, defined for the universal excitation field Ψ as:

$$L[\Psi] = \Psi^\dagger \nabla^2 \Psi - |\nabla \Psi|^2. \quad (2)$$

This operator measures whether an excitation forms a closed geometric boundary. Excitations satisfying $L[\Psi] = 0$ correspond to baryonic matter, analogous to Bogomolny-type solitonic structures [6]. Excitations for which $L[\Psi] \neq 0$ propagate through the internal $\text{SU}(3)$ manifold and do not form conserved geometric boundaries.

Using the amplitude–phase decomposition $\Psi = Ae^{i\theta}$, the operator becomes:

$$L[\Psi] = A \nabla^2 A - |\nabla A|^2 - A^2 |\nabla \theta|^2, \quad (3)$$

revealing contributions from amplitude curvature, gradient energy, and phase winding. These components are summarized in the Lemniscate Operator Chart in the Reference Charts section.

2.3 Conformal–Disformal Metric Structure

The dark-sector field Φ_{dark} modifies geometry through a combined conformal and disformal transformation. The conformal factor $k(\Phi_{\text{dark}})$ rescales the reference metric, while the disformal term $h(\Phi_{\text{dark}})$ introduces anisotropic stretching along gradients of Φ_{dark} .

This structure allows the dark sector to influence curvature, geodesic deviation, and matter propagation without introducing explicit new degrees of freedom in the metric tensor. The transformation is consistent with the general class of disformal metrics studied in [4,5].

2.4 Internal SU(3) Manifold

Neutrino flavor transport occurs on an internal SU(3) manifold whose geometric properties—curvature, torsion, and twist—define the geometric Hamiltonian H_{geom} . In this framework, the SU(3) manifold is treated as a classical internal geometry whose curvature, torsion, and twist encode effective flavor dynamics; quantum behavior enters only through the interpretation of these geometric quantities in oscillation experiments. The manifold is equipped with the following structures:

$$A = A_{\mu}^a \lambda_a dx^{\mu}, \tag{4}$$

$$F = dA + A \wedge A, \tag{5}$$

$$T^a = de^a + \omega^a_b \wedge e^b, \tag{6}$$

$$\Omega = \epsilon_{abc} e^a \wedge de^b. \tag{7}$$

Here:

- A is the SU(3) connection,
- F is the curvature 2-form,
- T^a is the torsion tensor,
- Ω is the twist (helicity) 2-form.

These operators appear throughout the geometric Hamiltonian and flavor evolution equations. Their definitions and physical interpretations are summarized in the Geometric Operators Chart in the Reference Charts section.

2.5 SU(3) Generators

The SU(3) generators λ_a play a central role in mapping geometric quantities to observable neutrino parameters. In particular:

- λ_3 corresponds to the atmospheric mass-splitting direction,
- λ_2 corresponds to the solar mixing direction,
- λ_8 governs global flavor-phase structure.

This mapping is detailed in the SU(3) Generator Chart in the Reference Charts section and is essential for connecting geometric corrections to PMNS-relevant observables.

2.6 Field Interaction Structure

The three fields interact through a causal chain:

$$\Phi_{\text{sub}} \rightarrow \rho_{\text{sub}} \rightarrow \Phi_{\text{dark}} \rightarrow g_{\mu\nu}^{\text{dark}} \rightarrow g_{\mu\nu} \rightarrow \Psi \rightarrow \text{SU(3) transport}.$$

This chain is summarized in the Field Coupling Chain Chart in the Reference Charts section and forms the backbone of the emergent-geometry framework.

2.7 Summary

This section establishes the mathematical and geometric structures used throughout the paper. The operators, metrics, and SU(3) quantities defined here appear in all subsequent sections, including the derivation of the geometric Hamiltonian, the mapping to PMNS parameters, and the analysis of observational signatures. Readers are encouraged to consult the Reference Charts section for a consolidated overview of these foundational elements.

3 Substrate Field Φ_{sub}

The substrate field Φ_{sub} forms the foundational layer of the emergent- geometry framework. It encodes proto-geometric density, seeds filamentary structure, and provides the initial geometric scaffolding upon which the dark-sector phase field, the foliation operator $\mathcal{F}$, and

the universal excitation field Ψ build. This section develops the manifold structure, action, density functional, gradient-flow dynamics, and coupling mechanisms associated with Φ_{sub} .

3.1 Manifold and Proto-Geometric Structure

We begin by defining a smooth manifold S without intrinsic geometric structure. The manifold is equipped only with the reference metric g_0 , which serves as a variational baseline but carries no physical dynamics. All geometric features—curvature, torsion, twist, foliation structure, and metric response—emerge from the fields defined on S .

The substrate field $\Phi_{\text{sub}} : S \rightarrow \mathbb{R}$ is a scalar field representing proto-geometric density. Its spatial gradients determine the initial alignment and organization of geometric features. In particular, the integral curves of $\nabla\Phi_{\text{sub}}$ form filamentary networks reminiscent of cosmic-web structure [3]. These filaments serve as geometric pre-structures that influence the behavior of the dark-sector field, the foliation operator $\mathcal{F}$, and the emergent metric.

3.2 Action and Field Dynamics

The dynamics of the substrate field are governed by the action

$$S_{\text{sub}} = \int d^4x \sqrt{-g_0} \left[\frac{1}{2} g_0^{\mu\nu} \partial_\mu \Phi_{\text{sub}} \partial_\nu \Phi_{\text{sub}} - U(\Phi_{\text{sub}}) \right], \quad (8)$$

where $U(\Phi_{\text{sub}})$ is a potential controlling the distribution and stability of substrate density. The form of U is left general, allowing for a wide range of proto-geometric behaviors, including domain formation, filament stabilization, and symmetry-breaking mechanisms that seed initial anisotropy. In particular, symmetry-breaking minima of U determine the initial anisotropic pattern that seeds filament formation.

Variation of the action yields the field equation

$$\nabla_0^2 \Phi_{\text{sub}} = \frac{dU}{d\Phi_{\text{sub}}}, \quad (9)$$

where ∇_0^2 is the Laplacian associated with the reference metric g_0 . Although g_0 carries no physical dynamics, it provides the differential structure necessary for defining field equations at the substrate level.

3.3 Substrate Density Functional

The substrate density ρ_{sub} is defined as a functional of the field and its derivatives:

$$\rho_{\text{sub}} = G(\Phi_{\text{sub}}, \partial\Phi_{\text{sub}}). \quad (10)$$

The functional G may incorporate:

- local density contributions from Φ_{sub} ,
- gradient contributions from $\partial\Phi_{\text{sub}}$,
- higher-order corrections from $\nabla^2\Phi_{\text{sub}}$.

The density functional plays a crucial role in the model:

1. It determines the expectation value of the dark-sector phase field $\langle\Phi_{\text{dark}}\rangle = f(\rho_{\text{sub}})$, providing the mechanism by which substrate gradients bias the intermediate metric $g_{\mu\nu}^{\text{dark}}$.
2. It influences the conformal and disformal response of the intermediate metric, linking substrate anisotropy to dark-sector anisotropy.
3. It sets the geometric environment in which the excitation field Ψ forms closed or open boundaries, thereby determining matter identity.
4. It provides the initial conditions for the foliation operator $\mathcal{F}$, whose level-set geometry responds to substrate density gradients.

These relationships are summarized in the Field Coupling Chain Chart in the Reference Charts section.

3.4 Filamentary Gradient Flow

One of the most important features of the substrate field is its ability to generate filamentary structure through gradient flow. Filaments correspond to integral curves of the gradient vector field $\nabla\Phi_{\text{sub}}$:

$$\dot{x}^i = \alpha \partial^i \Phi_{\text{sub}}, \quad (11)$$

where α is a flow parameter controlling the rate of filament formation.

These filaments serve as proto-geometric structures that influence:

- the alignment of dark-sector anisotropies,

- the organization of emergent curvature,
- the propagation environment for excitations,
- the initial conditions for SU(3) flavor transport,
- the sectoring structure of the foliation operator $\mathcal{F}$.

The characteristic filament alignment scale is given by

$$L_f \sim |\nabla \Phi_{\text{sub}}|^{-1}, \quad (12)$$

as summarized in the Geometric Quantities Chart in the Reference Charts section.

3.5 Coupling to Dark-Sector and Foliation Geometry

The substrate field influences both the dark-sector phase field and the foliation operator. The coupling mechanisms are:

Dark-sector coupling. Substrate density biases the dark-sector expectation value:

$$\langle \Phi_{\text{dark}} \rangle = f(\rho_{\text{sub}}), \quad (13)$$

which in turn modifies the intermediate metric $g_{\mu\nu}^{\text{dark}}$ through conformal–disformal response. This establishes a direct geometric link between substrate gradients and dark-sector anisotropy.

Foliation coupling. The foliation operator $\mathcal{F}$ responds to substrate gradients, aligning its level sets with filamentary structure. This coupling determines the geometry of the two-lobe sectoring (U_+ and U_-) and influences temporal orientation through the orientation field τ .

3.6 Geometric Interpretation

Although Φ_{sub} does not directly generate curvature, its gradient structure provides the initial geometric scaffolding upon which curvature emerges. The dark-sector field responds to substrate density, generating the intermediate metric $g_{\mu\nu}^{\text{dark}}$, which in turn influences the fully emergent metric $g_{\mu\nu}$. The foliation operator organizes this geometry into causally distinct sectors. Thus, the substrate field plays a foundational role in determining the large-scale geometric organization of spacetime.

3.7 Summary

The substrate field Φ_{sub} establishes the proto-geometric environment from which emergent geometry arises. Its gradient flow generates filamentary structure, its density functional influences dark-sector behavior, its coupling to $\mathcal{F}$ determines foliation-sector geometry, and its field equation governs the distribution of proto-geometric density. These features make Φ_{sub} the first and most fundamental component of the emergent-geometry framework.

4 Dark-Sector Phase Field Φ_{dark}

The dark-sector phase field Φ_{dark} constitutes the second layer of the three-field emergent-geometry framework. Whereas the substrate field Φ_{sub} establishes proto-geometric density and filamentary alignment, Φ_{dark} governs the intermediate geometric response through a combined conformal–disformal transformation. Its expectation value is determined by substrate density, and its gradients introduce anisotropic geometric bias that influences the foliation operator $\mathcal{F}$, the emergent metric $g_{\mu\nu}$, and the SU(3) geometric manifold governing neutrino flavor transport. This section develops the action, coupling structure, geometric role, and observational consequences of Φ_{dark} .

4.1 Action and Field Dynamics

The dark-sector field is modeled as a scalar field with canonical kinetic term and general potential:

$$S_{\text{dark}} = \int d^4x \sqrt{-g_0} \left[\frac{1}{2} g_0^{\mu\nu} \partial_\mu \Phi_{\text{dark}} \partial_\nu \Phi_{\text{dark}} - V(\Phi_{\text{dark}}) \right]. \quad (14)$$

The potential $V(\Phi_{\text{dark}})$ determines the phase structure of the dark sector. Depending on its form, the field may exhibit:

- slow-roll behavior reminiscent of dark-energy models,
- oscillatory dynamics producing phase-induced anisotropies,
- localized excitations contributing to structure formation,
- transitions between distinct dark-sector phases.

Variation of the action yields the field equation:

$$\nabla_0^2 \Phi_{\text{dark}} = \frac{dV}{d\Phi_{\text{dark}}}, \quad (15)$$

where ∇_0^2 is the Laplacian associated with the reference metric g_0 .

Although Φ_{dark} evolves on the reference manifold, its geometric impact is realized through the conformal–disformal transformation that generates the intermediate metric $g_{\mu\nu}^{\text{dark}}$.

4.2 Expectation Value Tracking and Substrate Coupling

The expectation value of the dark-sector field is determined by the substrate density:

$$\langle \Phi_{\text{dark}} \rangle = f(\rho_{\text{sub}}), \quad (16)$$

where f is a model-dependent function encoding the response of the dark sector to proto-geometric density.

This relationship establishes a direct geometric coupling chain:

1. Substrate density ρ_{sub} determines the local dark-sector phase.
2. The dark-sector phase determines the conformal factor $k(\Phi_{\text{dark}})$.
3. Gradients of Φ_{dark} determine the disformal term $h(\Phi_{\text{dark}})\partial_\mu\Phi_{\text{dark}}\partial_\nu\Phi_{\text{dark}}$.
4. Together, these generate the intermediate metric $g_{\mu\nu}^{\text{dark}}$.
5. The intermediate metric biases the foliation operator $\mathcal{F}$ and influences the emergent metric $g_{\mu\nu}$.

Thus, substrate gradients propagate upward through the geometric hierarchy via Φ_{dark} .

4.3 Conformal–Disformal Metric Response

The dark-sector field modifies geometry through a combined conformal and disformal transformation:

$$g_{\mu\nu}^{\text{dark}} = k(\Phi_{\text{dark}}) g_{0\mu\nu} + h(\Phi_{\text{dark}}) \partial_\mu\Phi_{\text{dark}} \partial_\nu\Phi_{\text{dark}}. \quad (17)$$

4.3.1 Conformal Factor

The conformal factor $k(\Phi_{\text{dark}})$ rescales the reference metric. It introduces isotropic modifications to geometric structure and can mimic the effects of dark energy or scalar–tensor theories.

4.3.2 Disformal Term

The disformal term

$$h(\Phi_{\text{dark}}) \partial_\mu \Phi_{\text{dark}} \partial_\nu \Phi_{\text{dark}}$$

introduces anisotropic stretching along gradients of Φ_{dark} . This term is responsible for:

- direction-dependent propagation effects,
- anisotropic structure formation,
- phase-induced CMB anisotropies,
- geometric biasing of excitation boundaries,
- sector-dependent foliation geometry.

The disformal structure follows the general class introduced by Bekenstein [4] and developed further in [5].

4.4 Intermediate Metric and Emergent Geometry

The intermediate metric $g_{\mu\nu}^{\text{dark}}$ plays a crucial role in the emergent-geometry hierarchy:

1. It mediates between substrate dynamics and the fully emergent metric.
2. It biases curvature formation by modifying geodesic structure.
3. It influences the propagation of the excitation field Ψ and the formation of closed boundaries.
4. It contributes to the geometric Hamiltonian through the coupling $g_{\mu\nu}^{\text{dark}} H_{\text{geom}}$.
5. It shapes the foliation operator $\mathcal{F}$, influencing temporal orientation and sectoring.

The intermediate metric is not itself the physical spacetime metric, but it shapes the emergent metric $g_{\mu\nu}$ through its influence on field interactions and geometric response.

4.5 Dark-Sector Anisotropies and Foliation Geometry

Oscillations or spatial variations in Φ_{dark} generate anisotropies in the intermediate metric. These anisotropies can produce observable signatures, including:

- CMB spectral distortions,
- directional variations in structure growth,
- alignment of large-scale structures with substrate filaments,
- deviations from isotropic expansion,
- sector-dependent anisotropy across U_+ and U_- .

The characteristic anisotropy scale is given by:

$$\ell_{\text{dark}} \sim \omega_{\text{dark}} D_{\text{LS}}, \quad (18)$$

where ω_{dark} is the oscillation frequency of Φ_{dark} and D_{LS} is the last-scattering distance.

These relationships are summarized in the Dark-Sector Response Chart in the Reference Charts section.

4.6 Role in Matter Identity and Neutrino Transport

The dark-sector field influences matter identity and neutrino transport indirectly:

- It modifies the geometric environment in which the boundary operator $L[\Psi]$ determines baryonic identity.
- It biases the $\text{SU}(3)$ geometric manifold through the coupling $g_{\mu\nu}^{\text{dark}} H_{\text{geom}}$, introducing foliation-dependent weighting in curvature, torsion, and twist.
- It contributes to geometric corrections in neutrino oscillation probabilities.
- It enforces foliation-compatible propagation by modifying the orientation field τ through anisotropic metric response.

Thus, Φ_{dark} plays a central role in linking substrate dynamics, emergent geometry, matter identity, foliation geometry, and flavor transport.

4.7 Summary

The dark-sector phase field Φ_{dark} provides the intermediate geometric response necessary for emergent spacetime structure. Through its conformal–disformal metric, it mediates between substrate density and the fully emergent metric, introduces anisotropies relevant to cosmology, shapes foliation-sector geometry, and influences matter identity and neutrino transport. These features make Φ_{dark} an essential component of the three-field emergent-geometry framework.

5 Universal Excitation Field Ψ

The universal excitation field Ψ forms the third and final component of the three-field emergent-geometry framework. Whereas the substrate field Φ_{sub} establishes proto-geometric density and filamentary alignment, and the dark-sector field Φ_{dark} generates the intermediate geometric response through conformal–disformal coupling, the excitation field Ψ determines matter identity and governs the transition between baryonic and non-baryonic behavior. Its boundary structure interacts with the foliation operator $\mathcal{F}$ and the temporal orientation field τ , linking emergent geometry to $\text{SU}(3)$ flavor transport. This section develops the amplitude–phase decomposition of Ψ , the geometric boundary operator $L[\Psi]$, the physical interpretation of closed versus open boundaries, and the role of Ψ in the geometric hierarchy.

5.1 Excitation Field Definition

The excitation field $\Psi : S \rightarrow \mathbb{C}$ is a complex scalar field representing universal excitations propagating on the emergent spacetime manifold. Its behavior determines whether an excitation forms a closed geometric boundary corresponding to baryonic matter or an open boundary corresponding to non-baryonic propagation through the internal $\text{SU}(3)$ manifold.

The field is decomposed into amplitude and phase:

$$\Psi(x) = A(x) e^{i\theta(x)}, \tag{19}$$

where A is a real amplitude and θ is a real phase. This decomposition allows the geometric boundary operator to be expressed in terms of amplitude curvature, gradient energy, and phase winding.

5.2 Geometric Boundary Operator

Matter identity is determined by the geometric boundary operator:

$$L[\Psi] = \Psi^\dagger \nabla^2 \Psi - |\nabla \Psi|^2. \quad (20)$$

Using the amplitude–phase decomposition, the operator becomes:

$$L[\Psi] = A \nabla^2 A - |\nabla A|^2 - A^2 |\nabla \theta|^2. \quad (21)$$

Each term has a clear geometric interpretation:

- $A \nabla^2 A$ measures amplitude curvature,
- $|\nabla A|^2$ measures gradient energy,
- $A^2 |\nabla \theta|^2$ measures phase winding.

These relationships are summarized in the Boundary Operator Chart in the Reference Charts section.

5.3 Closed Boundaries and Baryonic Identity

Excitations satisfying

$$L[\Psi] = 0 \quad (22)$$

form closed geometric boundaries. These boundaries behave analogously to topological solitons [6], exhibiting:

- conserved identity,
- localized structure,
- stability under geometric deformation,
- finite energy.

In the emergent-geometry framework, closed boundaries correspond to baryonic matter. Thus, baryonic identity is not imposed externally but arises from the geometric structure of the excitation field.

Closed boundaries interact with the foliation operator $\mathcal{F}$ by remaining confined to individual foliation leaves. This confinement ensures that baryonic matter respects the global temporal orientation defined by τ and does not cross the self-intersection locus Σ .

5.4 Open Boundaries and Non-Baryonic Propagation

Excitations for which

$$L[\Psi] \neq 0 \tag{23}$$

do not form closed boundaries. These excitations propagate through the internal $SU(3)$ manifold and do not exhibit conserved geometric identity. Neutrinos are the primary example of such excitations.

Open-boundary excitations experience:

- geometric flavor transport,
- curvature-induced phase accumulation,
- torsion-induced mixing,
- twist-induced helicity modulation.

Because open boundaries are not confined to foliation leaves, they propagate across the foliation structure in a manner constrained by the temporal orientation field τ . The condition $D\mathcal{F}(\dot{\gamma}) > 0$ ensures that non-baryonic excitations evolve monotonically across the foliation, enforcing orientation-preserving flavor transport.

5.5 Geometric Interpretation of Boundary Formation

The boundary operator $L[\Psi]$ determines whether an excitation forms a closed loop in the emergent geometry. This geometric interpretation provides a unified origin for matter identity:

- Closed boundaries correspond to baryonic matter.
- Open boundaries correspond to non-baryonic excitations.

The formation of closed boundaries depends on:

1. amplitude curvature $A\nabla^2 A$,
2. gradient energy $|\nabla A|^2$,
3. phase winding $A^2|\nabla\theta|^2$,
4. the geometric environment determined by $g_{\mu\nu}^{\text{dark}}$,
5. substrate alignment through $\nabla\Phi_{\text{sub}}$,

6. foliation-sector geometry through $\mathcal{F}$.

Thus, matter identity emerges from the interplay of all three fields and the foliation geometry.

5.6 Coupling to Emergent Geometry

The excitation field interacts with the emergent geometry through:

- the boundary operator $L[\Psi]$,
- the intermediate metric $g_{\mu\nu}^{\text{dark}}$,
- the fully emergent metric $g_{\mu\nu}$,
- the foliation operator $\mathcal{F}$,
- the temporal orientation field τ ,
- the internal $\text{SU}(3)$ manifold.

These interactions determine:

- whether an excitation forms a closed boundary,
- how excitations propagate through curved spacetime,
- how geometric quantities influence flavor evolution,
- how foliation-sector geometry influences non-baryonic transport.

5.7 Role in Neutrino Flavor Transport

For excitations with open boundaries, the excitation field couples to the internal $\text{SU}(3)$ manifold. The geometric Hamiltonian governing flavor evolution is:

$$H_{\text{geom}} = \alpha R \lambda_3 + \beta T \lambda_2 + \gamma \Omega \lambda_8, \quad (24)$$

where R , T , and Ω are curvature, torsion, and twist of the $\text{SU}(3)$ manifold. In this framework, the $\text{SU}(3)$ manifold is treated as a classical internal geometry whose curvature, torsion, and twist encode effective flavor dynamics.

The mapping between $\text{SU}(3)$ generators and PMNS-relevant directions is summarized in the $\text{SU}(3)$ Generator Chart in the Reference Charts section. Foliation constraints ensure that geometric-phase accumulation is monotonic across the self-intersection locus, producing sector-dependent oscillation signatures.

5.8 Summary

The universal excitation field Ψ determines matter identity through the geometric boundary operator $L[\Psi]$. Closed boundaries correspond to baryonic matter, while open boundaries correspond to non-baryonic excitations such as neutrinos. The field interacts with the emergent geometry through the intermediate metric, the foliation operator, the temporal orientation field, the fully emergent metric, and the internal $SU(3)$ manifold, providing a unified geometric origin for matter identity, temporal orientation, and flavor transport.

6 Lemniscate Foliation and Temporal Orientation

In the v1 formulation, we introduced the *lemniscate operator* $\mathcal{F} : M \rightarrow \mathbb{R}^2$, defined as a smooth map whose level sets reproduce the stratified lemniscate geometry central to our dual-sector cosmological model. In the present formulation, the operator plays a deeper role: its level-set structure induces a natural foliation of the manifold, and its self-intersection generates a global temporal orientation compatible with the geodesic evolution equations developed in Sections ??–??. The lemniscate operator thus provides the geometric backbone for the emergent arrow of time.

6.1 Definition of the Lemniscate Foliation

Let $\mathcal{F} = (\mathcal{F}_1, \mathcal{F}_2)$ denote the operator defined in v1. For each regular value $c \in \mathbb{R}^2$, the level set

$$\mathcal{F}_c = \mathcal{F}^{-1}(c)$$

defines a smooth one-dimensional leaf. The collection of all such leaves forms a foliation

$$\mathcal{F} = \{\mathcal{F}_c \mid c \in \mathbb{R}^2\},$$

which we refer to as the *lemniscate foliation* of M . The self-intersection of the classical lemniscate corresponds, in our operator formalism, to the set of critical values of $\mathcal{F}$ where the rank of $D\mathcal{F}$ drops. This critical set plays a distinguished geometric and dynamical role, acting as the pivot around which temporal orientation is organized.

6.2 Self-Intersection and Causal Pinching

The self-intersection of the lemniscate induces a stratification of the foliation into two lobes, corresponding to the two connected components of the regular-value domain. Let

$$\Sigma = \{x \in M \mid \text{rank}(D\mathcal{F}(x)) < 2\}$$

denote the self-intersection locus. Trajectories crossing Σ experience a *causal pinch*: the geodesic flow cannot reverse its orientation upon crossing, due to the compatibility conditions imposed by $\mathcal{F}$. This property is essential for the emergence of a global arrow of time. The pinch acts as a geometric diode, enforcing orientation-preserving flow across the foliation.

6.3 Temporal Orientation Field

We define a temporal orientation field

$$\tau : M \rightarrow TM$$

as a smooth, nowhere-vanishing vector field satisfying

$$D\mathcal{F}(\tau) > 0,$$

which ensures that τ is everywhere compatible with the orientation-preserving structure of the lemniscate operator. The condition $D\mathcal{F}(\tau) > 0$ enforces monotonicity of the $\mathcal{F}$ -coordinate along integral curves of τ , thereby establishing a global temporal orientation. The field τ provides the geometric mechanism by which local microdynamics—still symmetric under time reversal—assemble into a globally oriented temporal flow.

6.4 Arrow of Time from the Lemniscate Operator

The arrow of time emerges from the interplay between the foliation and the self-intersection locus. Let $\gamma(s)$ be a geodesic satisfying the evolution equations of Section ???. If $\gamma(s_0) \in \Sigma$, then the monotonicity condition

$$\frac{d}{ds}\mathcal{F}(\gamma(s)) > 0$$

implies that the geodesic cannot reverse its $\mathcal{F}$ -orientation after crossing Σ . This establishes a global, non-reversible temporal direction, even though the local microdynamics remain time-reversal symmetric. The lemniscate operator thus provides a geometric origin for temporal directionality, independent of entropy or statistical arguments.

6.5 Two-Lobe Cosmological Sectoring

The two lobes of the lemniscate correspond to distinct cosmological sectors. Let U_+ and U_- denote the connected components of $M \setminus \Sigma$. The curvature asymmetry built into the v1 definition of $\mathcal{F}$ induces distinct dynamical regimes on U_+ and U_- , corresponding respectively to expansion-dominated and contraction-dominated evolution. This dual-sector structure is a direct consequence of the operator’s geometry and is not present in generic lemniscate embeddings. The foliation therefore encodes a cosmological sectoring that influences geodesic flow, matter distribution, and large-scale anisotropy.

6.6 Simulation Considerations

The lemniscate foliation and temporal orientation field introduce nonlinear constraints on the geodesic flow and PDE evolution equations. Numerical simulation requires distributed computation to resolve the behavior near the self-intersection locus Σ , where the geometry is most sensitive. Parallel integration of geodesics, Monte Carlo sampling of foliation leaves, and surrogate modeling of the PDE dynamics are essential for exploring the full parameter space of the model. The foliation’s geometric constraints must be enforced at each integration step to preserve temporal orientation and causal consistency.

7 Internal SU(3) Geometry and Neutrino Transport

Non-baryonic excitations propagate through an internal SU(3) manifold whose geometric structure determines flavor evolution. Whereas baryonic matter arises from closed geometric boundaries determined by the operator $L[\Psi]$, excitations with open boundaries—most notably neutrinos—traverse an internal geometric space whose curvature, torsion, and twist define a flavor-transport Hamiltonian. In the present formulation, the SU(3) manifold is not an auxiliary construction but a geometric sector coupled to the lemniscate foliation and temporal orientation introduced in Section 6. The foliation imposes orientation-preserving constraints on propagation paths, ensuring compatibility between internal flavor transport and global temporal directionality.

7.1 Motivation for SU(3) Geometry

The choice of SU(3) as the internal manifold is motivated by several considerations:

- SU(3) is the smallest non-Abelian Lie group capable of encoding three-flavor mixing.

- Its eight generators λ_a provide a natural basis for mapping geometric quantities to PMNS-relevant directions.
- The manifold admits curvature, torsion, and twist, allowing geometric corrections to flavor evolution.
- SU(3) compactification appears in multiple contexts, including geometric-phase models [7] and topological approaches to flavor transport.
- The lemniscate foliation restricts admissible propagation paths, making SU(3) the minimal group compatible with orientation-preserving transport across foliation leaves.

The SU(3) Generator Chart in the Reference Charts section summarizes the mapping between generators and physical mixing directions.

7.2 Geometric Structures on the SU(3) Manifold

The SU(3) manifold is equipped with the following geometric structures:

7.2.1 Connection

$$A = A_\mu^a \lambda_a dx^\mu, \quad (25)$$

where A_μ^a are connection coefficients and λ_a are the Gell-Mann generators.

7.2.2 Curvature

$$F = dA + A \wedge A. \quad (26)$$

The curvature 2-form encodes geometric phase accumulation along propagation paths.

7.2.3 Torsion

$$T^a = de^a + \omega^a_b \wedge e^b, \quad (27)$$

where e^a are frame fields and ω^a_b is the spin connection. Torsion introduces mixing effects analogous to matter-induced flavor transitions.

7.2.4 Twist

$$\Omega = \epsilon_{abc} e^a \wedge de^b. \quad (28)$$

Twist (helicity) introduces energy-dependent phase modulation.

These operators are summarized in the Geometric Operators Chart in the Reference Charts section.

7.3 Foliation-Compatible Propagation

Propagation through the $SU(3)$ manifold must be compatible with the temporal orientation field τ defined in Section 6. Let $\gamma(s)$ be a propagation path. The compatibility condition

$$D\mathcal{F}(\dot{\gamma}) > 0 \quad (29)$$

ensures that flavor evolution respects the global arrow of time. This condition restricts admissible $SU(3)$ paths, preventing orientation-reversing loops and enforcing monotonic geometric-phase accumulation. The foliation therefore acts as a geometric constraint on internal flavor transport.

7.4 Geometric Hamiltonian

The geometric Hamiltonian governing flavor evolution is constructed from curvature, torsion, and twist:

$$H_{\text{geom}} = \alpha R \lambda_3 + \beta T \lambda_2 + \gamma \Omega \lambda_8, \quad (30)$$

where:

- R is curvature of the $SU(3)$ manifold,
- T is torsion,
- Ω is twist,
- λ_3 , λ_2 , and λ_8 are $SU(3)$ generators.

The coefficients (α, β, γ) determine the strength of geometric corrections and may depend on the emergent metric $g_{\mu\nu}$ or the intermediate metric $g_{\mu\nu}^{\text{dark}}$. The foliation imposes an additional constraint:

$$\frac{d}{ds} \mathcal{F}(\gamma(s)) > 0 \quad \Rightarrow \quad H_{\text{geom}} \text{ acts only on orientation-preserving paths.} \quad (31)$$

7.5 Mapping to PMNS-Relevant Directions

The $SU(3)$ generators map naturally onto PMNS-relevant directions:

- λ_3 corresponds to the atmospheric mass-splitting direction Δm_{31}^2 ,
- λ_2 corresponds to the solar mixing angle θ_{12} ,
- λ_8 governs global flavor-phase structure, including δ_{CP} .

This mapping is summarized in the SU(3) Generator Chart in the Reference Charts section and provides a direct route to experimental tests.

7.6 Flavor Evolution Equation

Flavor evolution along a propagation path $\gamma(\tau)$ is governed by:

$$\psi(\tau) = \exp \left[-i \int_0^\tau H_{\text{geom}}(\gamma(s)) ds \right] \psi_0, \quad (32)$$

where ψ_0 is the initial flavor state.

The geometric Hamiltonian introduces corrections to standard oscillation probabilities, including:

- curvature-induced phase shifts,
- torsion-induced mixing,
- twist-induced helicity modulation,
- dark-sector-induced anisotropies,
- foliation-induced orientation constraints.

These corrections arise from the geometric structure of the SU(3) manifold and the orientation-preserving constraints imposed by the foliation. The resulting flavor evolution is sensitive to both local geometric features and global foliation-sector structure.

7.7 Coupling to the Intermediate Metric

The geometric Hamiltonian couples to the intermediate metric through:

$$H_{\text{total}} = H_{\text{SM}} + \alpha' g_{\mu\nu}^{\text{dark}} H_{\text{geom}}, \quad (33)$$

where H_{SM} is the standard-model Hamiltonian including vacuum and matter effects (MSW). The coupling introduces:

- density-dependent geometric corrections,
- anisotropic flavor evolution,
- potential deviations from the MSW effect,
- directional dependence in long-baseline experiments,
- foliation-induced sector asymmetry.

Because $g_{\mu\nu}^{\text{dark}}$ depends on substrate density and dark-sector phase gradients, the geometric Hamiltonian inherits anisotropies from both fields. This coupling provides a mechanism by which cosmological structure, dark-sector dynamics, and foliation geometry influence neutrino oscillation probabilities.

7.8 Geometric Phase Accumulation

The curvature 2-form F contributes to geometric phase accumulation:

$$\Delta\phi_{\text{geom}} = \int_{\Sigma} F, \quad (34)$$

where Σ is a surface bounded by the propagation path. This phase is analogous to Berry-phase contributions in quantum systems and may produce observable deviations in oscillation probabilities. The foliation ensures that geometric-phase accumulation is monotonic across the self-intersection locus, preventing orientation-reversing loops and enforcing global temporal consistency.

Geometric-phase accumulation is sensitive to:

- local curvature of the $\text{SU}(3)$ manifold,
- foliation-sector geometry,
- dark-sector anisotropies,
- substrate-induced filament alignment,
- the orientation field τ .

These dependencies provide multiple observational pathways for detecting geometric corrections to flavor evolution.

7.9 Torsion-Induced Mixing

Torsion introduces mixing effects independent of mass differences. These effects may be detectable in:

- solar neutrino experiments (JUNO, SNO+),
- atmospheric neutrino detectors (IceCube, KM3NeT),
- long-baseline experiments (DUNE, Hyper-K).

Torsion-induced mixing arises from the geometric structure of the $SU(3)$ manifold and is enhanced in regions where the foliation geometry produces strong orientation gradients. Because torsion is sensitive to the alignment of frame fields, it provides a direct probe of geometric anisotropy in the internal manifold.

7.10 Twist-Induced Helicity Modulation

Twist introduces energy-dependent helicity modulation, potentially affecting:

- high-energy astrophysical neutrinos,
- neutrino flavor ratios observed at Earth,
- propagation through anisotropic dark-sector environments.

Twist-induced modulation arises from the helicity structure of the $SU(3)$ manifold and is amplified in regions where the dark-sector field exhibits strong gradients. This effect provides a geometric mechanism for energy-dependent deviations from standard oscillation predictions.

7.11 Sector-Dependent Flavor Evolution

Because the foliation divides spacetime into two lobes U_+ and U_- , flavor evolution may differ between sectors. The geometric Hamiltonian inherits sector asymmetry from:

- foliation geometry,
- dark-sector anisotropies,
- substrate filament alignment,
- orientation-field gradients.

Sector-dependent flavor evolution provides a potential explanation for directional anisotropies in astrophysical neutrino observations and may produce detectable signatures in long-baseline experiments.

7.12 Propagation Across the Self-Intersection Locus

Propagation across the self-intersection locus Σ is constrained by the orientation-preserving condition:

$$D\mathcal{F}(\dot{\gamma}) > 0. \quad (35)$$

This condition ensures that:

- geometric-phase accumulation is monotonic,
- torsion-induced mixing does not reverse orientation,
- twist-induced helicity modulation remains consistent across sectors,
- flavor evolution respects the global arrow of time.

Propagation across Σ therefore provides a geometric mechanism for enforcing global temporal orientation in flavor evolution.

7.13 Geometric Interpretation

The $SU(3)$ manifold provides a geometric origin for neutrino flavor transport. Its curvature, torsion, and twist define a geometric Hamiltonian that modifies oscillation probabilities, introduces geometric phases, and links flavor evolution to emergent geometry. The foliation imposes orientation-preserving constraints on propagation, ensuring compatibility between internal flavor transport and the global arrow of time.

The mapping between $SU(3)$ generators and PMNS-relevant directions provides a direct route to experimental tests, making the $SU(3)$ manifold a central component of the emergent-geometry framework.

7.14 Summary

The internal $SU(3)$ manifold provides a geometric origin for neutrino flavor transport. Curvature, torsion, and twist define a geometric Hamiltonian that modifies oscillation probabilities, introduces geometric phases, and links flavor evolution to emergent geometry. The foliation imposes orientation-preserving constraints on propagation, ensuring compatibility between

internal flavor transport and the global arrow of time. The mapping between $SU(3)$ generators and PMNS-relevant directions provides a direct route to experimental tests, making the $SU(3)$ manifold a central component of the emergent-geometry framework.

8 Mapping $SU(3)$ Generators to PMNS Directions

The $SU(3)$ generators provide a natural geometric basis for projecting curvature, torsion, and twist onto PMNS-relevant directions. In the emergent-geometry framework, this mapping is not imposed but arises from the structure of the internal manifold and its compatibility with the lemniscate foliation introduced in Section 6. The foliation restricts admissible propagation paths, ensuring that geometric-phase accumulation along $SU(3)$ directions respects the global temporal orientation. The $SU(3)$ Generator Chart in the Reference Charts section summarizes the correspondence between generators and physical mixing directions.

8.1 λ_3 : Atmospheric Sector

The generator λ_3 corresponds to the atmospheric mass-splitting direction Δm_{31}^2 . Curvature R projected onto λ_3 introduces:

- baseline-dependent geometric phase shifts,
- energy-dependent corrections to oscillation probabilities,
- potential deviations in long-baseline experiments (DUNE, Hyper-K).

Because curvature accumulation is monotonic across the foliation, atmospheric-sector corrections inherit orientation-preserving constraints from the lemniscate geometry.

8.2 λ_2 : Solar Sector

The generator λ_2 corresponds to the solar mixing angle θ_{12} . Torsion T projected onto λ_2 introduces:

- mixing-angle modulation,
- density-dependent geometric corrections,
- potential deviations in solar neutrino experiments (JUNO, SNO+).

Torsion couples directly to the intermediate metric $g_{\mu\nu}^{\text{dark}}$, producing anisotropic corrections that depend on the foliation sector (U_+ or U_-) traversed by the propagation path.

8.3 λ_8 : Global Flavor-Phase Structure

The generator λ_8 governs global flavor-phase structure, including the CP-violating phase δ_{CP} . Twist Ω projected onto λ_8 introduces:

- energy-dependent phase modulation,
- helicity-dependent corrections,
- potential deviations in atmospheric and astrophysical neutrino observations.

Because twist is sensitive to foliation-induced anisotropy, λ_8 corrections provide a geometric route to directional CP-phase variation.

8.4 Total Hamiltonian

The total Hamiltonian governing flavor evolution is:

$$H_{\text{total}} = H_{\text{vac}} + H_{\text{MSW}} + \alpha' g_{\mu\nu}^{\text{dark}} H_{\text{geom}}. \quad (36)$$

Coupling to the intermediate metric introduces:

- anisotropic geometric corrections,
- density-dependent geometric phases,
- directional dependence in flavor evolution,
- foliation-induced sector asymmetry.

8.5 Geometric Corrections to Oscillation Probabilities

Geometric corrections modify oscillation probabilities:

$$P_{\alpha \rightarrow \beta} = |\langle \nu_\beta | \exp(-iH_{\text{total}}L) | \nu_\alpha \rangle|^2. \quad (37)$$

Corrections include:

- curvature-induced phase shifts,
- torsion-induced mixing,
- twist-induced helicity modulation,

- dark-sector-induced anisotropies,
- foliation-induced orientation constraints.

These corrections may be detectable in:

- long-baseline experiments (DUNE, Hyper-K),
- solar neutrino detectors (JUNO, SNO+),
- atmospheric neutrino observatories (IceCube, KM3NeT),
- astrophysical neutrino measurements.

8.6 Experimental Signatures

The $SU(3)$ –PMNS correspondence provides several experimental signatures:

- deviations from standard oscillation probabilities,
- energy-dependent geometric phases,
- anisotropic flavor evolution,
- directional dependence in long-baseline experiments,
- potential deviations from the MSW effect,
- foliation-sector asymmetry in flavor ratios.

These signatures are summarized in the Hamiltonian Components and Experimental Probes Chart in the Reference Charts section.

8.7 Summary

The mapping between $SU(3)$ geometric directions and PMNS mixing parameters provides a direct bridge between internal manifold geometry and observable neutrino physics. Curvature, torsion, and twist project onto the atmospheric, solar, and global flavor-phase sectors through the generators $(\lambda_3, \lambda_2, \lambda_8)$, introducing geometric corrections to oscillation probabilities. Foliation-induced orientation constraints ensure that geometric-phase accumulation is monotonic across the self-intersection locus, producing sector-dependent signatures. The $SU(3)$ –PMNS correspondence therefore offers concrete experimental pathways for testing the geometric structure of the emergent-geometry framework across solar, atmospheric, long-baseline, and astrophysical neutrino observations.

9 Relation to Existing $SU(3)$ Compactification and Geometric Phase Models

The internal $SU(3)$ manifold introduced in this framework provides a geometric origin for neutrino flavor transport, linking curvature, torsion, and twist to observable mixing parameters. Although $SU(3)$ structures appear in several areas of theoretical physics—including compactification schemes, geometric-phase approaches, and topological models—the present construction differs in both motivation and implementation. This section situates the emergent-geometry $SU(3)$ manifold within the broader landscape of $SU(3)$ -based frameworks, highlighting similarities, differences, and novel features.

9.1 Overview of $SU(3)$ in Theoretical Physics

$SU(3)$ appears in multiple contexts:

- **Quantum Chromodynamics (QCD):** The color gauge group of the strong interaction.
- **Flavor $SU(3)$:** An approximate symmetry relating hadronic states.
- **Geometric-phase models:** $SU(3)$ manifolds used to describe Berry-phase accumulation in three-level systems [7].
- **Compactification schemes:** Higher-dimensional theories compactified on $SU(3)$ holonomy manifolds.
- **Topological soliton models:** $SU(3)$ target spaces for Skyrmion-like excitations.

The $SU(3)$ manifold in the emergent-geometry framework is distinct from these applications. It is not a gauge group, not a compactification space, and not a symmetry of the Lagrangian. Instead, it is an *internal geometric space* through which non-baryonic excitations propagate.

9.2 Comparison with $SU(3)$ Compactification Models

Compactification models typically involve:

- higher-dimensional manifolds,
- $SU(3)$ holonomy (e.g., Calabi–Yau spaces),

- geometric constraints on supersymmetry,
- moduli fields controlling compactification radii.

The present $SU(3)$ manifold differs in several key respects:

1. **Dimensionality:** It is an internal three-dimensional geometric space, not a six-dimensional holonomy manifold.
2. **Role in the theory:** It governs flavor transport, not compactification.
3. **Field dependence:** Its geometric properties (curvature, torsion, twist) depend on the emergent metric and the dark-sector field.
4. **Absence of moduli:** There are no moduli fields controlling the size or shape of the $SU(3)$ manifold.
5. **No supersymmetry constraints:** The manifold is not required to preserve supersymmetry or satisfy holonomy conditions.

Thus, while $SU(3)$ appears in compactification contexts, the present construction is conceptually distinct.

9.3 Comparison with Geometric-Phase Models

Geometric-phase models describe Berry-phase accumulation in three-level quantum systems. In the context of neutrino physics, Akhmedov [7] showed that geometric phases can arise in flavor evolution. These models typically involve:

- adiabatic evolution,
- cyclic parameter variation,
- Berry curvature on parameter space,
- $SU(3)$ structure in the space of Hamiltonians.

The emergent-geometry $SU(3)$ manifold differs in several ways:

1. **Geometric origin:** Curvature, torsion, and twist arise from the internal manifold itself, not from parameter-space evolution.
2. **Non-adiabatic behavior:** Flavor evolution need not be adiabatic; geometric corrections arise from the manifold geometry, not from cyclic evolution.

3. **Coupling to emergent geometry:** The geometric Hamiltonian couples to the intermediate metric $g_{\mu\nu}^{\text{dark}}$, introducing density-dependent and anisotropic corrections.
4. **Physical interpretation:** Geometric quantities correspond directly to PMNS-relevant directions through the mapping $(\lambda_3, \lambda_2, \lambda_8)$.

Thus, while geometric phases appear in both frameworks, the underlying mechanisms are different.

9.4 Comparison with Topological SU(3) Models

Topological models involving SU(3) target spaces typically describe:

- Skyrmion-like excitations,
- topological charge,
- soliton stability,
- baryon number conservation.

The present SU(3) manifold differs in several respects:

1. **Matter identity mechanism:** Baryonic identity arises from the boundary operator $L[\Psi]$, not from topological charge.
2. **Excitation behavior:** Non-baryonic excitations propagate through SU(3), whereas baryonic excitations form closed boundaries and do not.
3. **Geometric transport:** Flavor evolution is governed by curvature, torsion, and twist, not by topological charge conservation.
4. **Absence of soliton structure:** The SU(3) manifold does not host solitons; soliton-like behavior arises from Ψ .

Thus, the SU(3) manifold is not a topological target space.

9.5 Novel Features of the Present SU(3) Construction

The SU(3) manifold in the emergent-geometry framework introduces several novel features:

- **Geometric origin of flavor transport:** Flavor evolution arises from internal geometry, not from mass differences alone.

- **Mapping to PMNS parameters:** Curvature, torsion, and twist map directly onto atmospheric, solar, and global flavor-phase sectors.
- **Coupling to emergent geometry:** The geometric Hamiltonian couples to the intermediate metric, introducing density-dependent and anisotropic corrections.
- **Unified origin:** Flavor transport, matter identity, and emergent geometry arise from a common three-field substrate.
- **Experimental accessibility:** Geometric corrections may be observable in solar, atmospheric, long-baseline, and astrophysical neutrino experiments.

These features distinguish the present $SU(3)$ manifold from existing $SU(3)$ -based frameworks.

9.6 Summary

The internal $SU(3)$ manifold introduced in the emergent-geometry framework is distinct from $SU(3)$ structures appearing in compactification models, geometric-phase approaches, and topological soliton theories. Its role is to provide a geometric origin for neutrino flavor transport, with curvature, torsion, and twist mapping directly onto PMNS-relevant directions. The coupling to the intermediate metric introduces density-dependent and anisotropic corrections, offering novel experimental signatures and a unified geometric origin for flavor evolution.

10 Unified Action and Field Equations

The emergent-geometry framework integrates three interacting fields— Φ_{sub} , Φ_{dark} , and Ψ —together with the lemniscate foliation $\mathcal{F}$ and temporal orientation field τ . Their combined dynamics generate the fully emergent metric $g_{\mu\nu}$, determine matter identity through boundary formation, and govern neutrino flavor transport through an internal $SU(3)$ manifold. This section develops the unified action, derives the field equations, and presents the stress-energy tensors associated with each field. The resulting structure demonstrates how geometry, matter identity, temporal orientation, and flavor evolution arise from a common substrate.

10.1 Structure of the Unified Action

The total action is composed of seven parts:

$$S = S_{\text{grav}} + S_{\text{sub}} + S_{\text{dark}} + S_{\Psi} + S_{\text{Lem}} + S_{\tau} + S_{\nu}. \quad (38)$$

Each term plays a distinct role in the emergent-geometry framework.

10.1.1 Gravitational Action

The gravitational action is defined using the fully emergent metric $g_{\mu\nu}$:

$$S_{\text{grav}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} R[g], \quad (39)$$

where $R[g]$ is the Ricci scalar constructed from $g_{\mu\nu}$. Unlike traditional approaches, $g_{\mu\nu}$ is not fundamental; it emerges from the combined influence of the substrate, dark-sector, excitation fields, and foliation geometry.

10.1.2 Substrate Field Action

$$S_{\text{sub}} = \int d^4x \sqrt{-g_0} \left[\frac{1}{2} g_0^{\mu\nu} \partial_\mu \Phi_{\text{sub}} \partial_\nu \Phi_{\text{sub}} - U(\Phi_{\text{sub}}) \right]. \quad (40)$$

This action governs proto-geometric density and filamentary alignment.

10.1.3 Dark-Sector Field Action

$$S_{\text{dark}} = \int d^4x \sqrt{-g_0} \left[\frac{1}{2} g_0^{\mu\nu} \partial_\mu \Phi_{\text{dark}} \partial_\nu \Phi_{\text{dark}} - V(\Phi_{\text{dark}}) \right]. \quad (41)$$

The dark-sector field generates the intermediate conformal–disformal metric $g_{\mu\nu}^{\text{dark}}$.

10.1.4 Excitation Field Action

$$S_{\Psi} = \int d^4x \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \Psi^\dagger \partial_\nu \Psi - W(|\Psi|) \right]. \quad (42)$$

10.1.5 Boundary Operator Action

$$S_{\text{Lem}} = \int d^4x \sqrt{-g} \lambda_{\text{Lem}} L[\Psi], \quad (43)$$

where $L[\Psi]$ determines whether an excitation forms a closed geometric boundary.

10.1.6 Temporal Orientation Action

The temporal orientation field enforces monotonic flow across the foliation:

$$S_\tau = \int d^4x \sqrt{-g} \lambda_\tau D\mathcal{F}(\tau), \quad (44)$$

where the condition $D\mathcal{F}(\tau) > 0$ ensures compatibility with the global arrow of time.

10.1.7 Neutrino SU(3) Transport Action

$$S_\nu = \int d^4x \sqrt{-g} \psi^\dagger (i\partial_t - H_{\text{SM}} - \alpha' g_{\mu\nu}^{\text{dark}} H_{\text{geom}}) \psi. \quad (45)$$

This action incorporates:

- standard-model vacuum and matter effects,
- geometric corrections from curvature, torsion, and twist,
- foliation-induced orientation constraints,
- dark-sector-induced anisotropies.

10.2 Emergent Metric Construction

The fully emergent metric $g_{\mu\nu}$ is constructed from:

1. substrate density ρ_{sub} ,
2. dark-sector phase Φ_{dark} ,
3. excitation boundary structure $L[\Psi]$,
4. foliation geometry $\mathcal{F}$ and temporal orientation τ .

The intermediate metric $g_{\mu\nu}^{\text{dark}}$ biases the emergent metric through:

$$g_{\mu\nu} = g_{\mu\nu}^{\text{dark}} + \Delta g_{\mu\nu}[\Phi_{\text{sub}}, \Psi, \mathcal{F}, \tau]. \quad (46)$$

The correction term incorporates:

- filamentary alignment from $\nabla\Phi_{\text{sub}}$,
- boundary-induced curvature from $L[\Psi]$,
- foliation-induced anisotropy from $\mathcal{F}$,
- orientation-preserving constraints from τ .

10.3 Field Equations

Variation of the unified action yields the field equations.

10.3.1 Substrate Field Equation

$$\nabla_0^2 \Phi_{\text{sub}} = \frac{dU}{d\Phi_{\text{sub}}}. \quad (47)$$

10.3.2 Dark-Sector Field Equation

$$\nabla_0^2 \Phi_{\text{dark}} = \frac{dV}{d\Phi_{\text{dark}}}. \quad (48)$$

10.3.3 Excitation Field Equation

$$\nabla^2 \Psi - \frac{dW}{d\Psi^\dagger} + \lambda_{\text{Lem}} \frac{\delta L[\Psi]}{\delta \Psi^\dagger} = 0. \quad (49)$$

10.3.4 Boundary Operator Variation

$$\frac{\delta L[\Psi]}{\delta \Psi^\dagger} = \nabla^2 \Psi - \frac{\nabla \Psi \cdot \nabla \Psi^\dagger}{\Psi^\dagger}. \quad (50)$$

10.3.5 Temporal Orientation Equation

Variation of S_τ yields:

$$D\mathcal{F}(\tau) > 0, \quad (51)$$

enforcing monotonic flow across the foliation.

10.3.6 Neutrino Flavor Evolution Equation

$$i\partial_t \psi = (H_{\text{SM}} + \alpha' g_{\mu\nu}^{\text{dark}} H_{\text{geom}}) \psi. \quad (52)$$

10.4 Stress–Energy Tensors

Each field contributes to the emergent metric through its stress–energy tensor.

10.4.1 Substrate Stress–Energy Tensor

$$T_{\mu\nu}^{\text{sub}} = \partial_\mu \Phi_{\text{sub}} \partial_\nu \Phi_{\text{sub}} - g_{0\mu\nu} \left[\frac{1}{2} (\partial \Phi_{\text{sub}})^2 - U(\Phi_{\text{sub}}) \right]. \quad (53)$$

10.4.2 Dark-Sector Stress–Energy Tensor

$$T_{\mu\nu}^{\text{dark}} = \partial_\mu \Phi_{\text{dark}} \partial_\nu \Phi_{\text{dark}} - g_{0\mu\nu} \left[\frac{1}{2} (\partial \Phi_{\text{dark}})^2 - V(\Phi_{\text{dark}}) \right]. \quad (54)$$

10.4.3 Excitation Stress–Energy Tensor

$$T_{\mu\nu}^{\Psi} = \partial_{\mu}\Psi^{\dagger}\partial_{\nu}\Psi + \partial_{\nu}\Psi^{\dagger}\partial_{\mu}\Psi - g_{\mu\nu} \left[\frac{1}{2}(\partial\Psi)^2 - W(|\Psi|) \right]. \quad (55)$$

10.5 Summary

The unified action integrates substrate dynamics, dark-sector phase response, excitation boundary formation, foliation geometry, temporal orientation, and SU(3) flavor transport into a single geometric framework. The resulting field equations demonstrate how curvature, matter identity, temporal directionality, and flavor evolution arise from a common substrate. The stress–energy tensors of the three fields, together with foliation-induced anisotropy, collectively generate the fully emergent metric, linking geometric structure to observable phenomena across cosmology, particle physics, and neutrino oscillation experiments.

11 Observational Signatures

A central strength of the emergent-geometry framework is that it produces concrete, testable predictions across multiple observational domains. Because the model links substrate dynamics, dark-sector phase behavior, excitation boundary formation, lemniscate foliation geometry, temporal orientation, and SU(3) flavor transport, it yields signatures in cosmology, astrophysics, particle physics, and neutrino oscillation experiments. This section develops these observational consequences in detail, emphasizing how each geometric component contributes to measurable phenomena and how the combined framework can be distinguished from standard cosmology, modified gravity, and conventional neutrino physics.

11.1 Overview of Observable Channels

The model produces signatures in four broad categories:

1. **Cosmic structure formation** via substrate gradients, foliation-sectoring, and dark-sector disformal coupling.
2. **Cosmic microwave background anisotropies** via dark-sector phase oscillations, foliation-induced anisotropy, and intermediate metric response.
3. **Neutrino oscillation experiments** via geometric corrections from curvature, torsion, and twist of the SU(3) manifold, constrained by foliation orientation.

4. **Astrophysical neutrino propagation** via energy-dependent geometric phases, helicity modulation, and sector-dependent transport across the foliation.

These signatures are summarized in the Geometric Quantities Chart and the Hamiltonian Components and Experimental Probes Chart in the Reference Charts section.

11.2 Signatures from Substrate Field Φ_{sub}

The substrate field generates filamentary structure through gradient flow:

$$\dot{x}^i = \alpha \partial^i \Phi_{\text{sub}}. \quad (56)$$

This produces several observable consequences.

11.2.1 Filament Alignment Scale

The characteristic filament alignment scale is:

$$L_f \sim |\nabla \Phi_{\text{sub}}|^{-1}. \quad (57)$$

This scale may be detectable in:

- galaxy filament surveys (DESI, Euclid),
- weak-lensing maps (LSST),
- cosmic-web topology analyses.

Because filament alignment is influenced by foliation-sector geometry, alignment statistics may differ between the U_+ and U_- lobes, producing sector-dependent anisotropy.

11.2.2 Filament–CMB Correlations

If substrate gradients influence dark-sector anisotropies, filament orientations may correlate with CMB temperature or polarization anisotropies. Foliation-sectoring introduces a natural two-lobe asymmetry, providing a geometric explanation for hemispherical power anomalies or dipole-like features.

11.3 Signatures from Dark-Sector Phase Field Φ_{dark}

The dark-sector field generates the intermediate conformal–disformal metric:

$$g_{\mu\nu}^{\text{dark}} = k(\Phi_{\text{dark}}) g_{0\mu\nu} + h(\Phi_{\text{dark}}) \partial_\mu \Phi_{\text{dark}} \partial_\nu \Phi_{\text{dark}}. \quad (58)$$

This structure produces several observational signatures.

11.3.1 CMB Anisotropies

Oscillations in Φ_{dark} generate anisotropies in the intermediate metric, leading to:

- spectral distortions,
- directional anisotropies,
- potential deviations from isotropic expansion.

Because the foliation imposes a two-lobe sectoring, anisotropies may differ between U_+ and U_- , producing sector-dependent CMB signatures.

11.3.2 Modified Growth Factor

The disformal term modifies structure formation:

$$h(\Phi_{\text{dark}}) \partial_\mu \Phi_{\text{dark}} \partial_\nu \Phi_{\text{dark}} \quad \Rightarrow \quad \text{direction-dependent growth.} \quad (59)$$

This may produce:

- anisotropic growth rates,
- deviations in redshift-space distortions,
- scale-dependent clustering.

Sector-dependent growth asymmetry provides a geometric alternative to modified gravity.

11.4 Signatures from Excitation Field Ψ

The excitation field determines matter identity through the boundary operator:

$$L[\Psi] = A \nabla^2 A - |\nabla A|^2 - A^2 |\nabla \theta|^2. \quad (60)$$

11.4.1 Boundary-Induced Curvature

Closed boundaries ($L[\Psi] = 0$) generate localized curvature contributions to the emergent metric. These contributions may produce:

- small-scale deviations in gravitational lensing,
- localized curvature anomalies,
- potential signatures in strong-lensing arcs.

Because boundary formation is constrained by foliation geometry, curvature anomalies may cluster near the self-intersection locus Σ .

11.4.2 Matter Identity Transitions

Regions where $L[\Psi]$ transitions between zero and nonzero may produce:

- boundary-induced geometric defects,
- localized anisotropies,
- potential signatures in baryon–dark-matter offsets.

Sector-dependent transitions may produce asymmetric distributions across U_+ and U_- .

11.5 Signatures from SU(3) Geometric Flavor Transport

The geometric Hamiltonian introduces corrections to neutrino oscillation probabilities:

$$H_{\text{geom}} = \alpha R \lambda_3 + \beta T \lambda_2 + \gamma \Omega \lambda_8. \quad (61)$$

Because propagation must satisfy the foliation constraint $D\mathcal{F}(\dot{\gamma}) > 0$, geometric-phase accumulation is monotonic across the self-intersection locus, producing sector-dependent oscillation signatures.

11.5.1 Long-Baseline Experiments

Curvature, torsion, and twist may produce:

- baseline-dependent phase shifts,
- deviations from standard oscillation probabilities,

- potential anomalies in CP-violation measurements.

Sector-dependent corrections may produce directional asymmetries in DUNE and Hyper-K.

11.5.2 Solar Neutrino Experiments

Torsion-induced mixing may produce:

- deviations from the MSW effect,
- energy-dependent mixing-angle modulation,
- potential anomalies in day–night asymmetry.

Because torsion couples to $g_{\mu\nu}^{\text{dark}}$, solar-sector corrections may depend on dark-sector anisotropy.

11.5.3 Atmospheric Neutrino Observatories

Twist-induced helicity modulation may produce:

- energy-dependent flavor ratios,
- deviations in ν_μ/ν_e ratios,
- potential anomalies in upward-going neutrino fluxes.

Sector-dependent twist may produce hemispherical asymmetries in IceCube and KM3NeT.

11.6 Astrophysical Neutrino Signatures

High-energy astrophysical neutrinos provide a unique probe of geometric flavor transport.

11.6.1 Energy-Dependent Geometric Phases

Geometric phases may accumulate over cosmological distances:

$$\Delta\phi_{\text{geom}} = \int_{\Sigma} F, \tag{62}$$

leading to:

- energy-dependent flavor ratios,

- deviations from standard 1 : 1 : 1 expectations,
- potential anomalies in Glashow-resonance events.

Because propagation crosses foliation leaves, geometric-phase accumulation may differ between U_+ and U_- .

11.6.2 Directional Dependence

If geometric corrections couple to the intermediate metric, flavor ratios may depend on:

- source direction,
- dark-sector anisotropies,
- substrate filament alignment,
- foliation-sector geometry.

11.7 Combined Multi-Messenger Signatures

The model predicts correlations across observational channels:

- filament alignment $\leftrightarrow$ Cmb anisotropies,
- dark-sector oscillations $\leftrightarrow$ neutrino flavor anomalies,
- boundary-induced curvature $\leftrightarrow$ lensing deviations,
- $SU(3)$ geometric phases $\leftrightarrow$ astrophysical flavor ratios,
- foliation-sectoring $\leftrightarrow$ hemispherical asymmetries.

These correlations provide powerful tests of the emergent-geometry framework.

11.8 Summary

The emergent-geometry framework produces a rich set of observational signatures across cosmology, astrophysics, and neutrino physics. Substrate gradients influence filament structure, dark-sector phase oscillations modify the intermediate metric, excitation boundaries generate localized curvature, foliation geometry introduces sector-dependent anisotropy, and $SU(3)$ geometric transport introduces flavor-dependent corrections. These signatures provide concrete experimental pathways for testing the model and distinguishing it from standard cosmology, modified gravity, and conventional neutrino oscillation physics.

12 Conclusion

The emergent-geometry framework developed in this work unifies substrate dynamics, dark-sector phase response, foliation geometry, matter identity, and SU(3) flavor transport into a single coherent structure. Beginning from a manifold without intrinsic geometry, the model demonstrates how curvature, torsion, twist, temporal orientation, and matter identity arise from the collective behavior of three interacting fields: the substrate field Φ_{sub} , the dark-sector phase field Φ_{dark} , and the universal excitation field Ψ .

The substrate field establishes proto-geometric density and filamentary alignment, providing the initial scaffolding for geometric organization. The dark-sector field responds to substrate density through conformal–disformal coupling, generating the intermediate metric $g_{\mu\nu}^{\text{dark}}$ and introducing anisotropic geometric bias. The foliation operator $\mathcal{F}$ organizes this geometry into a stratified two-lobe structure, whose self-intersection locus Σ produces a global arrow of time through the temporal orientation field τ .

Matter identity emerges from the excitation field through the geometric boundary operator $L[\Psi]$. Closed boundaries correspond to baryonic matter, while open boundaries propagate through the internal SU(3) manifold. The SU(3) geometric space provides a natural environment for non-baryonic flavor transport, with curvature, torsion, and twist projecting onto the atmospheric, solar, and global flavor-phase sectors of the PMNS matrix. This correspondence yields concrete experimental predictions for long-baseline, solar, atmospheric, and astrophysical neutrino observations.

The unified action integrates all components of the framework, demonstrating how the emergent metric $g_{\mu\nu}$ arises from the combined influence of substrate density, dark-sector phase, excitation boundaries, and foliation geometry. The resulting field equations reveal a tightly coupled system in which geometric structure, matter identity, and flavor evolution are inseparable aspects of a single underlying mechanism.

The model produces a rich set of observational signatures across cosmology, astrophysics, and particle physics. Filament alignment, dark-sector anisotropies, boundary-induced curvature, and SU(3) geometric phases provide multiple pathways for testing the framework. Sector-dependent effects arising from foliation geometry offer distinctive predictions not present in standard cosmology, modified gravity, or conventional neutrino physics.

Several avenues for future work remain open. These include exploring symmetry-breaking mechanisms for initial substrate conditions, analyzing gravitational-wave propagation across foliation sectors, extending the SU(3) manifold to include higher-dimensional geometric phases, and developing numerical simulations capable of resolving behavior near the self-intersection locus. Further investigation of baryon number stability, dark-sector phase tran-

sitions, and multi-messenger correlations may reveal additional observable consequences.

Taken together, the results presented here show that emergent geometry can provide a unified origin for spacetime structure, matter identity, temporal orientation, and flavor evolution. The framework offers a new route toward connecting geometric foundations with experimental observables, opening the possibility that the structure of matter and time may ultimately arise from a single geometric substrate.

13 Reference Charts

This section consolidates the core structures of the emergent-geometry framework into a set of reference charts. These charts summarize the metric hierarchy, geometric operators, $SU(3)$ generator roles, field interactions, geometric quantities, Hamiltonian components, and experimental probes. They serve as a conceptual map for navigating the multi-layered structure of the model and provide a structural backbone for the framework.

13.1 Metric Hierarchy Chart

Metric / Structure	Symbol	Description
Reference Metric	g_0	Fixed background metric used for variational formulation. Carries no physical dynamics. Provides differential structure for substrate and dark-sector fields.
Intermediate Metric	$g_{\mu\nu}^{\text{dark}}$	Generated by conformal–disformal response of Φ_{dark} . Introduces anisotropic and density-dependent geometric modifications. Influences excitation propagation, foliation geometry, and geometric Hamiltonian.
Emergent Metric	$g_{\mu\nu}$	Fully dynamical spacetime metric arising from combined influence of Φ_{sub} , Φ_{dark} , Ψ , and foliation geometry. Governs curvature, torsion, and physical propagation.
Foliation Operator	$\mathcal{F}$	Defines foliation leaves $\mathcal{F}_c$, temporal orientation field τ , and self-intersection locus Σ . Introduces global causal structure and two-lobe cosmological sectoring (U_+, U_-) .

13.2 Boundary Operator Chart

Expression	$L[\Psi] = \Psi^\dagger \nabla^2 \Psi - \nabla \Psi ^2$
Amplitude–Phase Form	$L[\Psi] = A \nabla^2 A - \nabla A ^2 - A^2 \nabla \theta ^2$
Closed Boundary Condition	$L[\Psi] = 0$ corresponds to baryonic matter.
Open Boundary Condition	$L[\Psi] \neq 0$ corresponds to non-baryonic excitations (neutrinos).
Geometric Role	Determines whether an excitation forms a conserved geometric boundary. Controls matter identity and determines whether SU(3) transport is activated.
Foliation Link	Closed boundaries remain confined to foliation leaves $\mathcal{F}_c$; open boundaries propagate across leaves and obey temporal orientation constraints $D\mathcal{F}(\dot{\gamma}) > 0$.

13.3 Geometric Operators Chart

Operator	Symbol	Description
Connection	$A = A_\mu^a \lambda_a dx^\mu$	Defines parallel transport on SU(3) manifold.
Curvature	$F = dA + A \wedge A$	Encodes geometric phase accumulation. Projects onto λ_3 (atmospheric).
Torsion	$T^a = de^a + \omega^a_b \wedge e^b$	Introduces mixing independent of mass differences. Projects onto λ_2 (solar).
Twist (Helicity)	$\Omega = \epsilon_{abc} e^a \wedge de^b$	Introduces energy-dependent helicity modulation. Projects onto λ_8 (global flavor-phase).
Temporal Orientation Field	τ	Vector field satisfying $D\mathcal{F}(\tau) > 0$. Establishes global arrow of time and enforces monotonicity across Σ .

13.4 SU(3) Generator Chart

Generator	Symbol	Physical Role
Atmospheric Direction	λ_3	Maps curvature R onto atmospheric mass-splitting sector Δm_{31}^2 .
Solar Direction	λ_2	Maps torsion T onto solar mixing angle θ_{12} .
Global Flavor-Phase	λ_8	Maps twist Ω onto CP-phase structure δ_{CP} .

13.5 Field Coupling Chain Chart

Coupling Chain	$\Phi_{\text{sub}} \rightarrow \rho_{\text{sub}} \rightarrow \Phi_{\text{dark}} \rightarrow g_{\mu\nu}^{\text{dark}} \rightarrow g_{\mu\nu} \rightarrow \mathcal{F} \rightarrow \Psi \rightarrow \text{SU}(3) \text{ transport}$
Interpretation	Substrate density determines dark-sector phase. Dark-sector phase determines intermediate metric. Intermediate metric biases emergent geometry. Foliation operator imposes temporal orientation and sectoring. Excitation field determines matter identity. Open-boundary excitations propagate through SU(3).

13.6 Geometric Quantities Chart

Quantity	Symbol	Description
Filament Alignment Scale	$L_f \sim \nabla \Phi_{\text{sub}} ^{-1}$	Determines filament spacing and alignment. Observable in cosmic-web surveys.
Self-Intersection Locus	Σ	Critical set of $\mathcal{F}$ where rank drops. Generates causal pinch and non-reversible temporal orientation.
Foliation Leaf	$\mathcal{F}_c$	Level set of $\mathcal{F}$ defining one-dimensional geometric leaves.
Geometric Phase	$\Delta\phi_{\text{geom}} = \int_{\Sigma} F$	Curvature-induced phase accumulation. Observable in neutrino oscillations.

13.7 Hamiltonian Components Chart

Component	Symbol	Physical Role
Curvature Term	$\alpha R \lambda_3$	Atmospheric-sector geometric phase shifts.
Torsion Term	$\beta T \lambda_2$	Solar-sector mixing modulation.
Twist Term	$\gamma \Omega \lambda_8$	Global flavor-phase and helicity modulation.
Dark-Sector Coupling	$\alpha' g_{\mu\nu}^{\text{dark}} H_{\text{geom}}$	Density-dependent and anisotropic geometric corrections.

13.8 Experimental Probes Chart

Experiment Type	Observable Signature
Long-Baseline Neutrino Experiments	Baseline-dependent geometric phase shifts; deviations from standard oscillation probabilities; CP-phase anomalies; foliation-sector asymmetry.
Solar Neutrino Experiments	Torsion-induced mixing-angle modulation; deviations from MSW effect; dark-sector density dependence.
Atmospheric Neutrino Observatories	Energy-dependent flavor ratios; helicity modulation; hemispherical asymmetry from foliation-sector geometry.
Astrophysical Neutrino Telescopes	Cosmological geometric-phase accumulation; deviations from $1 : 1 : 1$ flavor ratios; directional dependence from dark-sector anisotropy.
Cosmic-Web Surveys	Filament alignment scale L_f ; anisotropic clustering; correlations with foliation-sector geometry.
CMB Observations	Dark-sector-induced anisotropies; polarization distortions; spectral deviations; sector-dependent signatures across U_+ and U_- .

13.9 Summary

These reference charts consolidate the core structures of the emergent-geometry framework, providing a visual and conceptual guide to the metric hierarchy, geometric operators, $SU(3)$ generator roles, field interactions, geometric quantities, Hamiltonian components, and experimental probes. They serve as a navigational tool for readers and a structural backbone for the theoretical model.

14 Bibliography

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